

however, the amplitude and phase of the scattered field are required; although this is not impossible in principle, it is rarely achieved in practice. The measurements usually available for analysis are the irradiance of the scattered light for a set of directions. We are therefore almost always faced with the task of trying to describe a particle (or worse yet, a collection of particles) with a less than theoretically ideal set of data in hand. But this is not necessarily cause for despair. Often, supplementary information about the particles, some of which is obtained by means other than light scattering techniques, is sufficient to enable them to be described. Thus, we should not be too ready to discard what little information is available or might become available. One source of information sometimes neglected is the polarization properties of the scattered light, a subject we treat in detail in succeeding chapters.

NOTES AND COMMENTS

An excellent concise treatment of scattering—by molecules and particles, single and multiple—at an intermediate level is Chapter 14 of Stone (1963). Among the books devoted entirely to scattering by particles, that by Shifrin (1951) most closely resembles ours in that it discusses optical properties of bulk matter as well. But the two books that have influenced us most are those of van de Hulst (1957) and Kerker (1969); we are indebted to both authors. Another book on scattering, which emphasizes polydispersions, is by Deirmendjian (1969).

There are also more limited treatments of scattering. McCartney (1976, Chaps. 4–6) confines his attention to scattering by atmospheric particles. This is also discussed by Twomey (1977, Chaps. 9–10) in his treatise on atmospheric aerosols. In Goody (1964, Chap. 7) there are discussions of absorption by gases and, in less detail, extinction by molecules and by droplets. Parts of books on electromagnetic theory or optics include the theory of scattering by a sphere, most notably Stratton (1941, pp. 563–573) and Born and Wolf (1965, pp. 633–664). The latter also derive the Ewald–Oseen extinction theorem and apply it to reflection and refraction at a plane interface (pp. 98–104).

The fluctuation theory of scattering by molecules is treated in books by Bhagavantam (1942), Fabelinskii (1968), and Chu (1974, Chap. 3).

Two of the more venerable works on multiple scattering are the review article by Milne (1930) and the book by Chandrasekhar (1950). A long-awaited treatise on this subject by van de Hulst (1980) recently made its appearance. Two reports, with many references, edited by Lenoble (1977) and by Fouquart et al. (1980), have been published by the International Association of Meteorology and Atmospheric Physics.

The inverse problem in scattering theory is discussed in the collections of papers edited by Baltes (1978, 1980).

Chapter 2

Electromagnetic Theory

The treatment of absorption and scattering of light by small particles is a problem in electromagnetic theory, a subdivision of which is optics. Various aspects of this theory are found scattered throughout books on electricity and magnetism, optics, and polarization of light. To rely on such existing books as the basis for our development might, however, result in a confusing tangle of conflicting assumptions, notation, and conventions. Also, it is convenient to gather together in one place the mathematical and physical apparatus that underlies succeeding chapters. Thus, we present in this chapter our version of those concepts and equations from electromagnetic theory that are germane to the subject at hand; we shall have need of them throughout the remainder of the book.

2.1 FIELD VECTORS AND THE MAXWELL EQUATIONS

As stated in the introductory chapter, we adopt a macroscopic approach to the problem of determining absorption and scattering of electromagnetic waves by particles. Therefore, the logical point of departure is the Maxwell equations for the macroscopic electromagnetic field at interior points in matter, which in SI units may be written

$$\nabla \cdot \mathbf{D} = \rho_f, \quad (2.1)$$

$$\nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} = 0, \quad (2.2)$$

$$\nabla \cdot \mathbf{B} = 0, \quad (2.3)$$

$$\nabla \times \mathbf{H} = \mathbf{J}_f + \frac{\partial \mathbf{D}}{\partial t}, \quad (2.4)$$

where $\mathbf{E}$ is the electric field and $\mathbf{B}$ the magnetic induction. The electric displacement $\mathbf{D}$ and magnetic field $\mathbf{H}$ are defined by

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}, \quad (2.5)$$

$$\mathbf{H} = \frac{\mathbf{B}}{\mu_0} - \mathbf{M}, \quad (2.6)$$

where $\mathbf{P}$ is the electric polarization (average electric dipole moment per unit volume), $\mathbf{M}$ the magnetization (average magnetic dipole moment per unit volume), ϵ_0 the permittivity, and μ_0 the permeability of free space. Implicit in (2.5) is the assumption that quadrupole and higher moments are negligible compared with the dipole moment. In free space the polarization and magnetization vanish identically. The charge density ρ_f and current density $\mathbf{J}_f$ are associated with so-called “free” charges. The terms “free” and “bound” are sometimes set in quotation marks, which indicates that they are slightly suspect. Indeed, Purcell (1963, pp. 342–347) has simply but convincingly shown that it is not always possible to unambiguously distinguish between free and bound charges in matter. Nevertheless, we shall assume in time-honored fashion that the ambiguity in the meanings of free and bound leads to no observable consequences in the problems with which we shall be concerned.

Although there appears to be nearly universal agreement about the microscopic Maxwell equations, it is not trivial to provide a derivation of the macroscopic equations that will satisfy everyone. The process of so doing was begun by Lorentz some 100 years ago, and new derivations appear regularly in the literature, each of which claims to be more general, freer from ambiguity, and more logically consistent than its predecessors. Without wishing to enter the lists of combatants, we direct the interested reader to Russakoff (1970) and Robinson (1973) for further discussion of the transition from the microscopic to the macroscopic field equations. We accept, therefore, (2.1)–(2.6) as *the* macroscopic field equations without further comment; any attempt to justify this choice rigorously and to define all terms precisely would lead to a volume that would dwarf the present work.

Equations (2.1)–(2.6) are not sufficient in themselves; they must be supplemented with *constitutive relations*, which are assumed to have the form

$$\mathbf{J}_f = \sigma \mathbf{E}, \quad (2.7)$$

$$\mathbf{B} = \mu \mathbf{H}, \quad (2.8)$$

$$\mathbf{P} = \epsilon_0 \chi \mathbf{E}, \quad (2.9)$$

where σ is the *conductivity*, μ the *permeability*, and χ the electric *susceptibility*. The *phenomenological coefficients* σ , μ , and χ depend on the medium under consideration, but will be assumed to be independent of the fields (the medium is *linear*), independent of position (the medium is *homogeneous*), and independent of direction (the medium is *isotropic*). There are many classes of materials for which these assumptions are *not* valid. Equations (2.7)–(2.9) are not universal laws of nature; they merely describe a particular class of materials which, fortunately, has a large number of members.

We shall not, however, assume that the phenomenological coefficients are independent of *frequency*. To provide some insight into this assertion, we must digress briefly. Consider, for example, the susceptibility. The polarization $\mathbf{P}$ is

the average dipole moment per unit volume of the medium, that is, the vector sum of the dipole moments in a unit volume. An isolated sample of matter (excluding electrets) is unpolarized ($\mathbf{P} = 0$). But when placed in an external field, which we may assume to be time harmonic, it becomes polarized: the electric field induces a net dipole moment. For a linear homogeneous isotropic medium, the relation (2.9) between $\mathbf{P}$ and $\mathbf{E}$ indicates that χ is a measure of how easily the material can be polarized; it represents the response of the material to the field $\mathbf{E}$. In particular, χ may be interpreted as the amplitude of the response to a unit field. Now it is well known (to those who know it well) that the response of a mechanical system to a periodic driving force is a sensitive function of the frequency. Therefore, by analogy, it is plausible that χ is a function of the frequency. The frequency dependence of χ is discussed in much greater detail in Chapters 9 and 10.

2.2 TIME-HARMONIC FIELDS

The general time-harmonic field $\mathbf{F}$ has the form

$$\mathbf{F} = \mathbf{A} \cos \omega t + \mathbf{B} \sin \omega t, \quad (2.10)$$

where ω is the *angular frequency*. The *real* vector fields $\mathbf{A}$ and $\mathbf{B}$ are independent of time but may depend on position. We note that $\mathbf{F}$ may be written as the real part of a complex vector: $\mathbf{F} = \text{Re}(\mathbf{F}_c)$, where

$$\mathbf{F}_c = \mathbf{C} \exp(-i\omega t), \quad \mathbf{C} = \mathbf{A} + i\mathbf{B}. \quad (2.11)$$

The vector $\mathbf{F}_c$ is a *complex representation* of the real field $\mathbf{F}$. If all our operations on time-harmonic fields are *linear* (e.g., addition, differentiation, integration), it is more convenient to work with the complex representation. The reason this may be done is as follows. Let $\mathcal{L}$ be any linear operator; we can operate on the field (2.10) by operating on the complex representation (2.11) and then take the real part of the result:

$$\mathcal{L}\mathbf{F} = \mathcal{L}\text{Re}(\mathbf{F}_c) = \text{Re}(\mathcal{L}\mathbf{F}_c).$$

Note that there is a degree of arbitrariness associated with the complex representation of a real field: $\mathbf{F}$ could just as easily have been written $\mathbf{F} = \text{Re}(\mathbf{F}_c^*)$, where $\mathbf{F}_c^* = \mathbf{C}^* \exp(i\omega t)$ and the asterisk denotes the complex conjugate. Thus, there are two possible choices for the time-dependent factor in a complex representation of a time-harmonic field: $\exp(i\omega t)$ and $\exp(-i\omega t)$. It makes no difference which choice is made: the quantities of physical interest are always real. But once a sign convention has been chosen it must be used consistently in all analysis. We shall take the time-dependent factor to be $\exp(-i\omega t)$; this is the convention found in standard books on optics (Born and Wolf, 1965) and electromagnetic theory (Stratton, 1941; Jackson, 1975) as

well as being nearly universal in solid-state physics. If comparison is made between our expressions and similar expressions in the scientific literature, the sign convention must be carefully noted.

If $\exp(-i\omega t)$ time dependence is assumed for all fields, and the constitutive relations (2.7)–(2.9) are substituted into (2.1)–(2.4), we obtain

$$\nabla \cdot (\epsilon \mathbf{E}_c) = 0, \quad (2.12)$$

$$\nabla \times \mathbf{E}_c = i\omega\mu\mathbf{H}_c, \quad (2.13)$$

$$\nabla \cdot \mathbf{H}_c = 0, \quad (2.14)$$

$$\nabla \times \mathbf{H}_c = -i\omega\epsilon\mathbf{E}_c, \quad (2.15)$$

where the complex *permittivity* is

$$\epsilon = \epsilon_0(1 + \chi) + i\frac{\sigma}{\omega}. \quad (2.16)$$

If $\epsilon \neq 0$, the electric field is divergence free; this is the general condition for a transverse field. Except possibly at frequencies where $\epsilon = 0$, therefore, the medium cannot support longitudinal fields.

Equations (2.12)–(2.16) will usually be our point of departure in scattering problems. However, to avoid a cluttered notation, we shall often omit the subscript c from the complex fields. In those instances where confusion might result, the subscript will be retained, although it should usually be clear from the context if we are dealing with real fields or their complex representations.

2.3 FREQUENCY-DEPENDENT PHENOMENOLOGICAL COEFFICIENTS

Although we gave some physical justification, the seemingly *ad hoc* introduction of frequency-dependent phenomenological coefficients implies that they have a well-defined meaning only if the fields are harmonic in time. But this is very unsatisfactory: it is certainly not a law of nature that all electromagnetic fields must be harmonic in time. Thus, we must answer the questions: How are $\mathbf{P}$ and $\mathbf{E}$ (or $\mathbf{B}$ and $\mathbf{H}$) related in general, and what is the meaning of frequency-dependent phenomenological coefficients? To answer these questions we must invoke the mathematical apparatus of Fourier transforms.

2.3.1 Fourier Transforms

Consider a *real* function of time $F(t)$. The Fourier transform $\mathcal{F}(\omega)$ of $F(t)$ is defined as

$$\mathcal{F}(\omega) = \int_{-\infty}^{\infty} F(t) e^{i\omega t} dt, \quad (2.17)$$

where ω is real. We assume that the integral in (2.17) converges in some sense for all functions of interest. The *inverse Fourier transform* is given by

$$F(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \mathcal{F}(\omega) e^{-i\omega t} d\omega. \quad (2.18)$$

According to (2.18), an arbitrary time-dependent function can be expressed as a superposition of time-harmonic functions $\exp(-i\omega t)$, where the complex amplitude $\mathcal{F}(\omega)$ depends on the frequency ω . The condition that $F(t)$ be real is that $\mathcal{F}^*(\omega) = \mathcal{F}(-\omega)$; therefore, $F(t)$ can be expressed as a superposition of time-harmonic functions with positive frequency:

$$F(t) = \text{Re}\{F_c(t)\},$$

$$F_c(t) = \frac{1}{\pi} \int_0^{\infty} \mathcal{F}(\omega) e^{-i\omega t} d\omega.$$

Note that (2.17) and (2.18) are not unique: all complex functions could be replaced by their complex conjugates, and the factor $1/2\pi$ could appear either in (2.17) or (2.18). If we want our expressions to appear more symmetrical, both integrals can have the common multiplicative factor $1/\sqrt{2\pi}$. There is no universally accepted convention for Fourier transforms. However, once the form of the Fourier transform has been specified, the corresponding expression for the inverse Fourier transform is uniquely determined.

If we Fourier analyze the Maxwell equations (2.1)–(2.4), with $\rho_f = 0$, and assume that the operations of integration and differentiation may be interchanged, we obtain

$$\nabla \cdot \mathcal{D}(\omega) = 0, \quad (2.19)$$

$$\nabla \cdot \mathcal{B}(\omega) = 0, \quad (2.20)$$

$$\nabla \times \mathcal{E}(\omega) = i\omega \mathcal{B}(\omega), \quad (2.21)$$

$$\nabla \times \mathcal{H}(\omega) = \mathcal{J}_f(\omega) - i\omega \mathcal{D}(\omega), \quad (2.22)$$

where

$$\mathcal{E}(\omega) = \int_{-\infty}^{\infty} \mathbf{E}(t) e^{i\omega t} dt; \quad \text{etc.}$$

The Fourier transforms of the fields are, in general, position dependent, although this is not explicitly indicated. Because the fields are real, we need consider only positive frequencies. Let us now assume that the Fourier transforms of the fields, rather than the fields themselves, are connected by

linear constitutive relations:

$$\mathcal{P}(\omega) = \epsilon_0 \chi(\omega) \mathcal{E}(\omega), \quad (2.23)$$

$$\mathcal{J}_f(\omega) = \sigma(\omega) \mathcal{E}(\omega), \quad (2.24)$$

$$\mathcal{B}(\omega) = \mu(\omega) \mathcal{H}(\omega), \quad (2.25)$$

where $\chi^*(\omega) = \chi(-\omega)$, and so on. We also assume that $\chi(\omega)$, $\sigma(\omega)$, and $\mu(\omega)$ are scalar quantities independent of position. As we shall see, the constitutive relations (2.23)–(2.25) reduce to familiar expressions when the fields are time harmonic but, in addition, answer the question posed above: How are arbitrarily time-dependent fields related?

Before we proceed we need one more result from Fourier transform theory. If the Fourier transform $\mathcal{X}(\omega)$ of a function $X(t)$ is expressed as a product of Fourier transforms

$$\mathcal{X}(\omega) = \mathcal{Y}(\omega) \mathcal{Z}(\omega),$$

the *convolution theorem* states that

$$X(t) = \int_{-\infty}^{\infty} Y(t-t') Z(t') dt',$$

where

$$Y(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \mathcal{Y}(\omega) e^{-i\omega t} d\omega; \quad \text{etc.}$$

If we apply the convolution theorem to the constitutive relation (2.23), for example, we obtain

$$\mathcal{P}(t) = \int_{-\infty}^{\infty} G(t-t') \mathcal{E}(t') dt', \quad (2.26)$$

where

$$G(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \epsilon_0 \chi(\omega) e^{-i\omega t} d\omega. \quad (2.27)$$

Thus, a frequency-dependent susceptibility $\chi(\omega)$ implies that the polarization $\mathcal{P}$ at time t depends on the electric field $\mathcal{E}$ at all other times t' . This conclusion is consistent with simple physical reasoning. If, for example, a steady electric field is applied to a sample of matter for a sufficient period of time, a steady polarization will be induced in the sample. However, if the electric field were to be suddenly removed, the polarization would not immediately drop to zero but would decay according to characteristic times associated with microscopic processes. In this example it is clear that the polarization is not proportional to the instantaneous field.

Let us now assume that the fields are time harmonic with angular frequency ω_0 :

$$\mathcal{E}(t) = \mathbf{A} \cos \omega_0 t + \mathbf{B} \sin \omega_0 t = \text{Re}\{\mathcal{E}_c\}.$$

The Fourier transform of $\mathbf{E}(t)$ is

$$\mathfrak{E}(\omega) = \pi\{(\mathbf{A} + i\mathbf{B})\delta(\omega - \omega_0) + (\mathbf{A} - i\mathbf{B})\delta(\omega + \omega_0)\},$$

where δ is the Dirac delta function. From the Fourier inversion formula (2.18) and the constitutive relation (2.23) it follows that

$$\begin{aligned} \mathbf{P}(t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \epsilon_0 \chi(\omega) \mathfrak{E}(\omega) e^{-i\omega t} d\omega \\ &= \frac{\epsilon_0}{2} \{(\mathbf{A} + i\mathbf{B})\chi(\omega_0) e^{-i\omega_0 t} + (\mathbf{A} - i\mathbf{B})\chi(-\omega_0) e^{i\omega_0 t}\}. \end{aligned}$$

If we use the relation $\chi(-\omega_0) = \chi^*(\omega_0)$, then

$$\mathbf{P}(t) = \text{Re}\{\mathbf{P}_c(t)\} = \text{Re}\{\epsilon_0 \chi(\omega_0) \mathbf{E}_c(t)\}.$$

Thus, the perhaps unfamiliar constitutive relations (2.23)–(2.25) yield familiar results when the fields are time harmonic; moreover, because of (2.26) and (2.27), physical meaning can now be attached to the phenomenological coefficients even for arbitrarily time-dependent fields.

Although we concentrated attention on the electric susceptibility, similar remarks hold for the conductivity and permeability. In particular, we note that μ can be frequency dependent and is not restricted to be real. Complex phenomenological coefficients imply differences in *phase* between the various time-harmonic fields. For example, if the permeability $\mu = \mu' + i\mu''$ is complex, then for a real magnetic field $\mathbf{H} = \mathbf{H}_0 \cos \omega t$, the corresponding magnetic induction $\mathbf{B}$ is $\bar{\mu} \mathbf{H}_0 \cos(\omega t - \phi)$, where the phase angle ϕ is given by $\tan \phi = \mu''/\mu'$ and $\bar{\mu} = \{(\mu')^2 + (\mu'')^2\}^{1/2}$. In Section 2.6 we shall show that if the imaginary part of any of the phenomenological coefficients of a medium is nonzero, the amplitude of a plane wave will decrease as it propagates through such a medium because of absorption of electromagnetic energy. Thus, complex phenomenological coefficients or, alternatively, a phase difference between $\mathbf{P}$ and $\mathbf{E}$ (or $\mathbf{B}$ and $\mathbf{H}$) are physically manifested by absorption, the detailed nature of which is hidden from view in a macroscopic approach.

If the Fourier-transformed Maxwell equations (2.19)–(2.22) and the constitutive relations (2.23)–(2.25) are combined, we obtain

$$\nabla \cdot \mathfrak{E}(\omega) = 0, \quad (2.28)$$

$$\nabla \cdot \mathfrak{H}(\omega) = 0, \quad (2.29)$$

$$\nabla \times \mathfrak{E}(\omega) = i\omega\mu(\omega)\mathfrak{H}(\omega), \quad (2.30)$$

$$\nabla \times \mathfrak{H}(\omega) = -i\omega\epsilon(\omega)\mathfrak{E}(\omega), \quad (2.31)$$

where $\epsilon(\omega) = \epsilon_0(1 + \chi(\omega)) + i\sigma(\omega)/\omega$. Note that the two sets of equations (2.12)–(2.15) and (2.28)–(2.31) are formally identical although interpreted differently. In the former set, $(\mathbf{E}_c, \mathbf{H}_c)$ is the complex representation of a time-harmonic field; whereas in the latter set, $(\mathfrak{E}, \mathfrak{H})$ is the Fourier transform of an arbitrarily time-dependent electromagnetic field. But the same equations have the same solutions, so we need consider only time-harmonic fields; the general time-dependent field can be constructed by Fourier synthesis.

Both the conductivity and the susceptibility contribute to the imaginary part of the permittivity: $\text{Im}\{\epsilon\} = \text{Im}\{\chi\} + \text{Re}\{\sigma/\omega\}$. A nonzero value for $\text{Im}\{\epsilon\}$ manifests itself physically by absorption of electromagnetic energy in the medium. We may associate $\text{Im}\{\chi\}$ with the “bound” charge current density and $\text{Re}\{\sigma/\omega\}$ with the “free” charge current density. Absorption is determined by the *sum* of these two quantities, however, and it is not possible to determine by absorption measurements their relative contributions. This underscores our assertion that there is no clearly defined distinction between “free” and “bound” charges.

2.3.2 Kramers–Kronig Relations

As an alternative we could have begun our discussion of constitutive relations with the *assumption* that $\mathbf{P}(t)$ and $\mathbf{E}(t)$ are related through the linear functional equation

$$\mathbf{P}(t) = \int_{-\infty}^{\infty} G(t, t') \mathbf{E}(t') dt'. \quad (2.32)$$

Suppose that the electric field is a delta function applied at time t_0 : $\mathbf{E}(t) = \delta(t - t_0)\mathbf{E}_0$; the corresponding polarization is therefore

$$\mathbf{P}(t) = G(t, t_0)\mathbf{E}_0.$$

Thus, G is the polarization resulting from a unit amplitude delta function. If the properties of the medium do not change with time, the polarization must depend only on the time elapsed between t_0 and t :

$$G(t, t_0) = G(t - t_0).$$

Therefore, we obtain (2.26), which when inverted yields the constitutive relation (2.23). We note that *causality*—the system cannot squeal before it is hurt—requires that $G(\tau) = 0$ for $\tau < 0$.

The susceptibility is the Fourier transform of $G(t)$:

$$\epsilon_0\chi(\omega) = \int_{-\infty}^{\infty} G(t)e^{i\omega t} dt = \int_0^{\infty} G(t)e^{i\omega t} dt, \quad (2.33)$$

and is a complex-valued function of the real variable ω . Let us define a

complex-valued function of the *complex* variable $\tilde{\omega}$ by

$$\epsilon_0 \chi(\tilde{\omega}) = \int_0^\infty G(t) e^{i\tilde{\omega}t} dt,$$

where $\tilde{\omega} = \omega_r + i\omega_i$. The function $\chi(\tilde{\omega})$ coincides with $\chi(\omega)$ when $\tilde{\omega}$ is a point on the real axis. For any $t \geq 0$, $G(t)e^{i\tilde{\omega}t}$ is an analytic function of $\tilde{\omega}$, and $|G(t)e^{i\tilde{\omega}t}| \leq |G(t)|$ if $\omega_i > 0$. Therefore, if the integral

$$\int_0^\infty |G(t)| dt \quad (2.34)$$

converges, then

$$\int_0^\infty G(t) e^{i\tilde{\omega}t} dt$$

converges to an analytic function in the upper half of the complex $\tilde{\omega}$ plane. Convergence of (2.34) is assured if $\chi(0)$ is finite; this follows from (2.33). If $\chi(\tilde{\omega})$ is analytic, so is $\chi(\tilde{\omega})/(\tilde{\omega} - \omega)$ except at the pole $\tilde{\omega} = \omega$. Cauchy's theorem states that

$$\int_C f(\tilde{\omega}) d\tilde{\omega} = 0,$$

provided that the closed contour C encloses no poles of the analytic function $f(\tilde{\omega})$. Let us apply Cauchy's theorem to the function $\chi(\tilde{\omega})/(\tilde{\omega} - \omega)$, where ω is a point on the real axis, and the contour C , shown in Fig. 2.1, is the union of four curves with parametric representations

$$C_1: \quad \tilde{\omega} = \Omega \quad (-A \leq \Omega \leq \omega - a),$$

$$C_2: \quad \tilde{\omega} = \omega - ae^{-i\Omega} \quad (0 \leq \Omega \leq \pi),$$

$$C_3: \quad \tilde{\omega} = \Omega \quad (\omega + a \leq \Omega \leq A),$$

$$C_4: \quad \tilde{\omega} = Ae^{i\Omega} \quad (0 \leq \Omega \leq \pi).$$

Therefore, from Cauchy's theorem

$$\begin{aligned} \int_{-A}^{\omega-a} \frac{\chi(\Omega)}{\Omega - \omega} d\Omega + \int_{\omega+a}^A \frac{\chi(\Omega)}{\Omega - \omega} d\Omega + \int_0^\pi \frac{iAe^{i\Omega}\chi(Ae^{i\Omega})}{Ae^{i\Omega} - \omega} d\Omega \\ = \int_0^\pi i\chi(\omega - ae^{i\Omega}) d\Omega. \end{aligned}$$

The integral over the curve C_4 vanishes as A tends to infinity if $\lim_{|\tilde{\omega}| \rightarrow \infty} \chi(\tilde{\omega})$

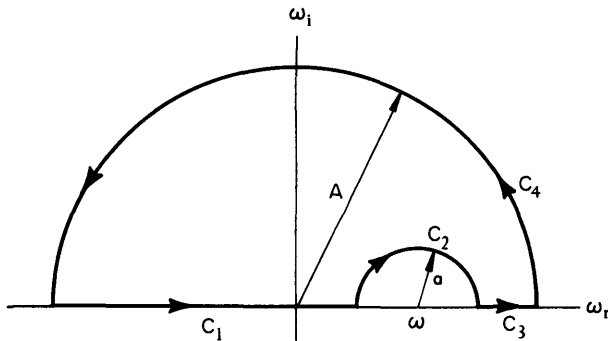


Figure 2.1 Contour of integration.

$= 0$; this implies that

$$i\pi\chi(\omega) = P \int_{-\infty}^{\infty} \frac{\chi(\Omega)}{\Omega - \omega} d\Omega, \quad (2.35)$$

where the symbol P denotes the *Cauchy principal value* of the integral, defined by

$$\lim_{a \rightarrow 0} \int_{-\infty}^{\omega-a} \frac{\chi(\Omega)}{\Omega - \omega} d\Omega + \int_{\omega+a}^{\infty} \frac{\chi(\Omega)}{\Omega - \omega} d\Omega.$$

We were slightly careless in our derivation of (2.35): $\chi(\tilde{\omega})$ was required to be analytic in the upper half of the complex $\tilde{\omega}$ plane, whereas part of the contour C included the real axis. But this can easily be remedied, and a more general form of (2.35) is

$$i\pi \lim_{\eta \rightarrow 0} \chi(\omega + i\eta) = \lim_{\eta \rightarrow 0} P \int_{-\infty}^{\infty} \frac{\chi(\Omega + i\eta)}{\Omega - \omega} d\Omega,$$

from which (2.35) follows if $\chi(\tilde{\omega})$ is continuous.

The fundamental relation (2.35) can be written as two real integral relations:

$$\chi'(\omega) = \frac{2}{\pi} P \int_0^{\infty} \frac{\Omega \chi''(\Omega)}{\Omega^2 - \omega^2} d\Omega, \quad (2.36)$$

$$\chi''(\omega) = \frac{-2\omega}{\pi} P \int_0^{\infty} \frac{\chi'(\Omega)}{\Omega^2 - \omega^2} d\Omega, \quad (2.37)$$

where $\chi = \chi' + i\chi''$ and we have invoked the *crossing condition* $\chi^*(\omega) = \chi(-\omega)$. Equations (2.36) and (2.37) are an important example of a rather

remarkable class of mathematical relations called *Kramers–Kronig* or *dispersion relations*. Their implication, which is unexpected on physical grounds and therefore has an aura of black magic, is that the real and imaginary parts of χ are not independent but are connected by integral relations; this imposes a constraint on physically realizable susceptibilities. Moreover, if χ' is known over a sufficiently large range of frequencies around ω , $\chi''(\omega)$ can be obtained by integration, and vice versa. An interesting corollary of (2.36) is the *sum rule*:

$$\chi'(0) = \frac{2}{\pi} \int_0^\infty \frac{\chi''(\Omega)}{\Omega} d\Omega.$$

Although the Kramers–Kronig relations do not follow directly from physical reasoning, they are not devoid of physical content: underlying their derivation are the assumptions of *linearity* and *causality* and restrictions on the asymptotic behavior of χ . As we shall see in Chapter 9, the required asymptotic behavior of χ is a physical consequence of the interaction of a frequency-dependent electric field with matter.

The derivation of Kramers–Kronig relations for the susceptibility was relatively easy, perhaps misleadingly so. With a bit of extra effort, however, we can often derive similar relations for other frequency-dependent quantities that arise in physical problems. Suppose that we have two time-dependent quantities of unspecified origin, which we may call the *input* $X_i(t)$ and the *output* $X_o(t)$; the corresponding Fourier transforms are denoted by $\mathcal{X}_i(\omega)$ and $\mathcal{X}_o(\omega)$. If the relation between these transforms is linear,

$$\mathcal{X}_o(\omega) = \mathcal{R}(\omega) \mathcal{X}_i(\omega),$$

and causal (i.e., the output cannot precede the input in time), then $\mathcal{R}(\tilde{\omega})$ is an analytic function in the top half of the complex $\tilde{\omega}$ plane. It is also necessary that $\mathcal{R}(\tilde{\omega})$ vanish on the circular arc C_4 (Fig. 2.1) as A approaches infinity; if it does not, we are permitted to fiddle with it until it does. That is, we can change the asymptotic behavior of $\mathcal{R}(\tilde{\omega})$ by multiplying it by some analytic function $g(\tilde{\omega})$, or adding $g(\tilde{\omega})$ to it, without changing its analyticity. Of course, in so doing, we may also change the crossing condition, and the resultant dispersion relations may be different from (2.36) and (2.37). Techniques of fiddling with $\mathcal{R}$ until it behaves properly are best illustrated with specific examples, which we shall encounter later in this chapter and elsewhere in the book.

2.4 SPATIAL DISPERSION

We have shown that a frequency-dependent susceptibility implies *temporal dispersion*: the polarization at time t depends on the electric field at all times previous to t . It is also possible under some circumstances to have *spatial dispersion*: the polarization at point $\mathbf{x}$ depends on the values of the electric field at points in some neighborhood of $\mathbf{x}$. This nonlocal relation between $\mathbf{P}$ and $\mathbf{E}$

leads to the concept of a susceptibility that depends on frequency *and* wave vector. To understand this it is helpful to introduce the three-dimensional Fourier transform of an arbitrary real function $F(\mathbf{x})$:

$$\mathcal{F}(\mathbf{k}) = \int F(\mathbf{x}) e^{-i\mathbf{k} \cdot \mathbf{x}} d\mathbf{x},$$

where the integration is taken over all space. The inverse Fourier transform is

$$F(\mathbf{x}) = \frac{1}{(2\pi)^3} \int \mathcal{F}(\mathbf{k}) e^{i\mathbf{k} \cdot \mathbf{x}} d\mathbf{k}.$$

Let us Fourier analyze the polarization and the electric field:

$$\mathcal{P}(\mathbf{k}, \omega) = \int \int \mathbf{P}(\mathbf{x}, t) e^{-i(\mathbf{k} \cdot \mathbf{x} - \omega t)} d\mathbf{x} dt,$$

$$\mathcal{E}(\mathbf{k}, \omega) = \int \int \mathbf{E}(\mathbf{x}, t) e^{-i(\mathbf{k} \cdot \mathbf{x} - \omega t)} d\mathbf{x} dt.$$

If we assume that there is a linear relation $\mathcal{P}(\mathbf{k}, \omega) = \epsilon_0 \chi(\mathbf{k}, \omega) \mathcal{E}(\mathbf{k}, \omega)$, the convolution theorem yields

$$\mathbf{P}(\mathbf{x}, t) = \int \int G(\mathbf{x} - \mathbf{x}', t - t') \mathbf{E}(\mathbf{x}', t') d\mathbf{x}' dt',$$

where

$$G(\mathbf{x}, t) = \frac{1}{(2\pi)^4} \int \int \epsilon_0 \chi(\mathbf{k}, \omega) e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)} d\mathbf{k} d\omega.$$

Thus, a susceptibility that depends on frequency *and* wave vector implies that the relation between $\mathbf{P}(\mathbf{x}, t)$ and $\mathbf{E}(\mathbf{x}, t)$ is nonlocal in time *and* space. Such spatially dispersive media lie outside our considerations. However, spatial dispersion can be important when the wavelength is comparable to some characteristic length in the medium (e.g., mean free path), and it is well at least to be aware of its existence: it can have an effect on absorption and scattering by small particles (Yildiz, 1963; Foley and Pattanayak, 1974; Ruppin, 1975, 1981).

2.5 POYNTING VECTOR

Consider an electromagnetic field $(\mathbf{E}, \mathbf{H})$, which is not necessarily time harmonic. The *Poynting vector* $\mathbf{S} = \mathbf{E} \times \mathbf{H}$ specifies the magnitude and direction of the rate of transfer of electromagnetic energy at all points of space; it is

of fundamental importance in problems of propagation, absorption, and scattering of electromagnetic waves. If the orientation of a plane surface with area A is specified by a unit normal vector $\hat{\mathbf{n}}$, the rate at which electromagnetic energy is transferred across this surface is $\mathbf{S} \cdot \hat{\mathbf{n}} A$, provided that $\mathbf{S}$ is constant over the surface. When $\mathbf{S}$ is a function of position and the surface has arbitrary shape, this can be generalized to

$$\int \mathbf{S} \cdot \hat{\mathbf{n}} dA.$$

The net rate W at which electromagnetic energy crosses the boundary of a closed surface A which encloses a volume V is

$$W = - \int_A \mathbf{S} \cdot \hat{\mathbf{n}} dA.$$

(Why the minus sign? There are two choices for the unit normal to a closed surface: inward and outward. We have chosen the outward normal. If $\mathbf{S}$ and $\hat{\mathbf{n}}$ are oppositely directed at a given point ($\mathbf{S} \cdot \hat{\mathbf{n}} < 0$), the minus sign ensures that the contribution to W is positive. Therefore, W is positive if there is a net transfer of electromagnetic energy *into* the volume. A positive W implies that electromagnetic energy is *absorbed* in V ; that is, electromagnetic energy is converted into other forms of energy (e.g., thermal energy) within V .)

(The formation of the vector product of two vectors is not a linear operation. Therefore, if the electromagnetic field is time harmonic, it is *not* true that $\mathbf{S} = \text{Re}(\mathbf{E}_c \times \mathbf{H}_c)$, although it is true that >

$$\mathbf{S} = \text{Re}\{\mathbf{E}_c\} \times \text{Re}\{\mathbf{H}_c\}. \quad (2.38)$$

The instantaneous Poynting vector (2.38) is a rapidly varying function of time for frequencies that are usually of interest. Most instruments are not capable of following the rapid oscillations of the instantaneous Poynting vector, but respond to some time average $\langle \mathbf{S} \rangle$:

$$\langle \mathbf{S} \rangle = \frac{1}{\tau} \int_t^{t+\tau} \mathbf{S}(t') dt',$$

where τ is a time interval long compared with $1/\omega$. The time-averaged Poynting vector for time-harmonic fields is given by

$$\langle \mathbf{S} \rangle = \frac{1}{2} \text{Re}\{\mathbf{E}_c \times \mathbf{H}_c^*\}.$$

When it is clear from the context that it is the time-averaged Poynting vector with which we are dealing, the brackets enclosing $\mathbf{S}$ will be omitted.

2.6 PLANE-WAVE PROPAGATION IN UNBOUNDED MEDIA

Let us “look for” plane-wave solutions to the Maxwell equations (2.12)–(2.15). What does this statement mean? We know that the electromagnetic field ($\mathbf{E}$, $\mathbf{H}$) cannot be arbitrarily specified. Only certain electromagnetic fields, those that satisfy the Maxwell equations, are physically realizable. Therefore, because of their simple form, we should like to know under what conditions plane electromagnetic waves

$$\mathbf{E}_c = \mathbf{E}_0 \exp(i\mathbf{k} \cdot \mathbf{x} - i\omega t); \quad \mathbf{H}_c = \mathbf{H}_0 \exp(i\mathbf{k} \cdot \mathbf{x} - i\omega t), \quad (2.39)$$

where $\mathbf{E}_0$ and $\mathbf{H}_0$ are constant vectors, are compatible with the Maxwell equations. The wave vector $\mathbf{k}$ may be complex

$$\mathbf{k} = \mathbf{k}' + i\mathbf{k}'', \quad (2.40)$$

where $\mathbf{k}'$ and $\mathbf{k}''$ are real vectors. If (2.40) is substituted in (2.39), we obtain

$$\mathbf{E}_c = \mathbf{E}_0 \exp(-\mathbf{k}'' \cdot \mathbf{x}) \exp(i\mathbf{k}' \cdot \mathbf{x} - i\omega t),$$

$$\mathbf{H}_c = \mathbf{H}_0 \exp(-\mathbf{k}'' \cdot \mathbf{x}) \exp(i\mathbf{k}' \cdot \mathbf{x} - i\omega t).$$

$\mathbf{E}_0 \exp(-\mathbf{k}'' \cdot \mathbf{x})$ and $\mathbf{H}_0 \exp(-\mathbf{k}'' \cdot \mathbf{x})$ are the *amplitudes* of the electric and magnetic waves, and $\phi = \mathbf{k}' \cdot \mathbf{x} - \omega t$ is the *phase* of the waves. An equation of the form $\mathbf{K} \cdot \mathbf{x} = \text{constant}$, where $\mathbf{K}$ is any real vector, defines a plane surface the normal to which is $\mathbf{K}$. Therefore, $\mathbf{k}'$ is perpendicular to the *surfaces of constant phase*, and $\mathbf{k}''$ is perpendicular to the *surfaces of constant amplitude*. If $\mathbf{k}'$ and $\mathbf{k}''$ are parallel, which includes the case $\mathbf{k}'' = 0$, these surfaces coincide and the waves are said to be *homogeneous*; if $\mathbf{k}'$ and $\mathbf{k}''$ are not parallel, the waves are said to be *inhomogeneous*. For example, waves propagating in a vacuum are homogeneous.

Let us briefly consider propagation of surfaces of constant phase. Choose an arbitrary origin O and a plane surface over which the phase ϕ is constant (Fig. 2.2). At time t the distance from the origin O to the plane is z , where $\mathbf{k}' \cdot \mathbf{x} = k'z$ and $k'z - \omega t = \phi$. In a time interval Δt the surface of constant phase will have moved a distance Δz , where

$$k'z - \omega t = k'(z + \Delta z) - \omega(t + \Delta t) = \phi.$$

Thus, the velocity of propagation of surfaces of constant phase, the *phase velocity* v , is

$$v = \frac{\Delta z}{\Delta t} = \frac{\omega}{k'},$$

and the vector $\mathbf{k}'$ specifies the direction of propagation.

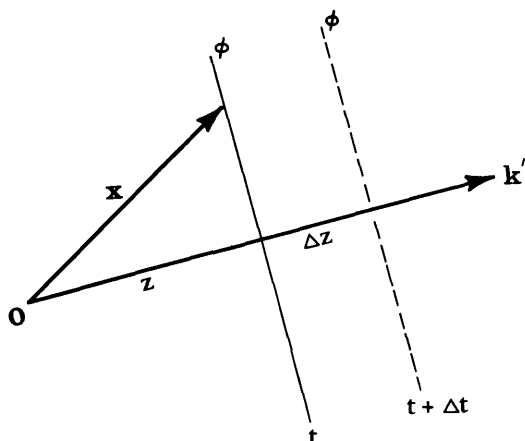


Figure 2.2 Propagation of constant phase surfaces.

The Maxwell equations for plane waves are

$$\mathbf{k} \cdot \mathbf{E}_0 = 0, \quad (2.41)$$

$$\mathbf{k} \cdot \mathbf{H}_0 = 0, \quad (2.42)$$

$$\mathbf{k} \times \mathbf{E}_0 = \omega \mu \mathbf{H}_0, \quad (2.43)$$

$$\mathbf{k} \times \mathbf{H}_0 = -\omega \epsilon \mathbf{E}_0. \quad (2.44)$$

Equations (2.41) and (2.42) are the conditions for *transversality*: $\mathbf{k}$ is perpendicular to $\mathbf{E}_0$ and $\mathbf{H}_0$. It is also evident from (2.43) or (2.44) that $\mathbf{E}_0$ and $\mathbf{H}_0$ are perpendicular. However, $\mathbf{k}$, $\mathbf{E}_0$, and $\mathbf{H}_0$ are, in general, complex vectors, and the interpretation of the term “perpendicular” is not simple unless the waves are homogeneous; for such waves, the real fields $\mathbf{E}$ and $\mathbf{H}$ lie in a plane the normal to which is parallel to the direction of propagation.

If we take the vector product of both sides of (2.43) with $\mathbf{k}$,

$$\mathbf{k} \times (\mathbf{k} \times \mathbf{E}_0) = \omega \mu \mathbf{k} \times \mathbf{H}_0 = -\omega^2 \epsilon \mu \mathbf{E}_0,$$

and use the **BAC–CAB** vector identity:

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$$

together with (2.41), we obtain

$$\mathbf{k} \cdot \mathbf{k} = \omega^2 \epsilon \mu. \quad (2.45)$$

Our analysis shows that plane waves (2.39) are compatible with the **Maxwell** equations provided that $\mathbf{k}$, $\mathbf{E}_0$, and $\mathbf{H}_0$ are perpendicular:

$$\mathbf{k} \cdot \mathbf{E}_0 = \mathbf{k} \cdot \mathbf{H}_0 = \mathbf{E}_0 \cdot \mathbf{H}_0 = 0.$$

In addition, the wave vector must satisfy (2.45); this equation may be written

$$k'^2 - k''^2 + 2i\mathbf{k}' \cdot \mathbf{k}'' = \omega^2 \epsilon \mu. \quad (2.46)$$

Note that $\epsilon\mu$ is a property of the *medium* in which the wave propagates; the vectors $\mathbf{k}'$ and $\mathbf{k}''$, however, are properties of the *wave*. If $\mathbf{k}'$ and $\mathbf{k}''$ satisfy (2.46), they may otherwise be arbitrary: ϵ and μ do not uniquely specify the details of wave propagation.

The wave vector of a homogeneous wave may be written $\mathbf{k} = (k' + ik'')\hat{\mathbf{e}}$, where k' and k'' are nonnegative and $\hat{\mathbf{e}}$ is a real unit vector in the direction of propagation. Equation (2.45) requires that

$$\mathbf{k} = \mathbf{k}' + i\mathbf{k}'' = \frac{\omega N}{c},$$

where c is the speed of light *in vacuo* and the *complex refractive index* N is

$$N = c\sqrt{\epsilon\mu} = \sqrt{\frac{\epsilon\mu}{\epsilon_0\mu_0}}. \quad (2.47)$$

We shall write

$$N = n + ik, \quad (2.48)$$

where n and k are nonnegative. Other notation commonly encountered for the complex refractive index is $n' + in''$ and $n(1 + i\kappa)$; if we had chosen $\exp(i\omega t)$ time dependence, these would be written $n' - in''$ and $n(1 - i\kappa)$. The symbol k , where $\omega\sqrt{\epsilon\mu} = k$, is widely used to denote the wave number. On the other hand, the use of k to denote the imaginary part of the complex refractive index is equally sacrosanct, particularly among those who actually undertake the task of its measurement. We shall adhere to both of these conventions; however, to differentiate between the two quantities, we shall use italic type for the imaginary part of the refractive index and Roman type for the wave number. The use of similar symbols to denote two physically distinct but not unrelated quantities may possibly lead to confusion; but even more befuddlement would undoubtedly result to many readers if we were to meddle with well-established conventions for the sake of symbolic precision.

The free-space wave number is $\omega/c = 2\pi/\lambda$, where λ is the wavelength *in vacuo*. Therefore, a plane homogeneous wave has the form

$$\mathbf{E}_c = \mathbf{E}_0 \exp\left(-\frac{2\pi k z}{\lambda}\right) \exp\left(\frac{i2\pi n z}{\lambda} - i\omega t\right),$$

where $z = \hat{\epsilon} \cdot \mathbf{x}$. Thus, the imaginary part of the complex refractive index determines the attenuation of the wave as it propagates through the medium; the real part determines the phase velocity $v = c/n$. The pair of quantities n and k are often referred to as the *optical constants*, terminology that is as widespread as it is misleading: the optical "constants" are not constant, they often strongly depend on frequency. It is not unusual for k of many common solids to range over six orders of magnitude within a relatively narrow range of frequencies.

Although refractive indices of many transparent materials at visible wavelengths have been accurately known for a long time, experimental determination of optical constants is by no means trivial in wavelength regions where a solid or liquid is appreciably absorbing. Special techniques have had to be developed to measure optical constants for various materials in different portions of the electromagnetic spectrum, and lack of reliable optical constants is still a serious impediment to small-particle scattering and absorption calculations in several areas of applied physics.

The real and imaginary parts of the complex refractive index satisfy Kramers-Kronig relations; sometimes this can be used to assess the reliability of measured optical constants. $N(\omega)$ satisfies the same crossing condition as $\chi(\omega)$: $N^*(\omega) = N(-\omega)$. However, it does not vanish in the limit of indefinitely large frequency: $\lim_{\omega \rightarrow \infty} N(\omega) = 1$. But this is a small hurdle, which can be surmounted readily enough by minor fiddling with $N(\omega)$: the quantity $N(\omega) - 1$ has the desired asymptotic behavior. If we now assume that $N(\tilde{\omega})$ is analytic in the top half of the complex $\tilde{\omega}$ plane, it follows that

$$n(\omega) - 1 = \frac{2}{\pi} P \int_0^\infty \frac{\Omega k(\Omega)}{\Omega^2 - \omega^2} d\Omega, \quad (2.49)$$

$$k(\omega) = \frac{-2\omega}{\pi} P \int_0^\infty \frac{n(\Omega)}{\Omega^2 - \omega^2} d\Omega, \quad (2.50)$$

where we used

$$P \int_0^\infty \frac{d\Omega}{\Omega^2 - \omega^2} = 0 \quad (2.51)$$

in the derivation of (2.50).

2.6.1 Absorption of Electromagnetic Energy

The Poynting vector of a plane wave is

$$\mathbf{S} = \frac{1}{2} \text{Re}(\mathbf{E} \times \mathbf{H}^*) = \text{Re} \left\{ \frac{\mathbf{E} \times (\mathbf{k}^* \times \mathbf{E}^*)}{2\omega\mu^*} \right\},$$

where $\mathbf{E} \times (\mathbf{k}^* \times \mathbf{E}^*) = \mathbf{k}^*(\mathbf{E} \cdot \mathbf{E}^*) - \mathbf{E}^*(\mathbf{k} \cdot \mathbf{E})$. If the wave is homogeneous, $\mathbf{k} \cdot \mathbf{E} = 0$ implies that $\mathbf{k}^* \cdot \mathbf{E} = 0$; for such a wave propagating in the $\hat{\epsilon}$

direction, we have

$$\mathbf{S} = \frac{1}{2} \operatorname{Re} \left\{ \sqrt{\frac{\epsilon}{\mu}} \right\} |\mathbf{E}_0|^2 \exp \left(-\frac{4\pi k z}{\lambda} \right) \hat{\mathbf{e}}.$$

Not surprisingly, $\mathbf{S}$ is in the direction of propagation. The magnitude of $\mathbf{S}$, which we shall denote by the symbol I , is called the *irradiance* and its dimensions are energy per unit area and time. (The term *intensity* is often used to denote irradiance; however, intensity is also used for other radiometric quantities, and we shall therefore tend to avoid this term because of possible confusion. E is now the recommended symbol for irradiance, but this hardly seems appropriate in a book where the electric field and irradiance often appear side by side.) As the wave traverses the medium, the irradiance is exponentially attenuated:

$$I = I_0 e^{-\alpha z},$$

where the *absorption coefficient* α is

$$\alpha = \frac{4\pi k}{\lambda} \quad (2.52)$$

and I_0 is the irradiance at $z = 0$. Throughout this chapter we adopt the viewpoint that bulk matter is homogeneous. However, this is only approximately true; even in media that are usually considered to be homogeneous, such as bulk samples of a pure liquid or solid, a beam of light is attenuated both by absorption and by scattering. Although absorption is usually the dominant mode of attenuation in such media, scattering is not entirely absent, and unless special techniques are used a measurement of attenuation unavoidably yields the combined effect of absorption and scattering. In later chapters we shall discuss attenuation by collections of material particles; such attenuation may or may not be dominated by absorption depending on the size and optical properties of the particles.

The rate at which electromagnetic energy is removed from the wave as it propagates through the medium is determined by the imaginary part of the complex refractive index. If the irradiances I_0 and I_t (or rather their ratio) are measured at two different positions $z = 0$ and $z = h$, then α , and hence k , can be obtained in principle from the relation

$$\alpha h = \ln \frac{I_0}{I_t}. \quad (2.53)$$

This equation is strictly valid only if the detector is optically identical with the medium for which the absorption coefficient is to be measured (Fig. 2.3a), a condition that is difficult to satisfy. The usual experimental configuration is

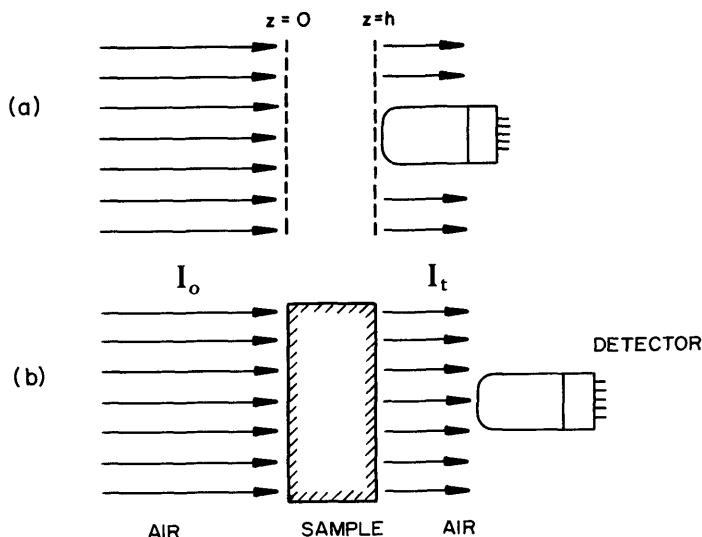


Figure 2.3 Measurement of absorption: (a) in principle and (b) in practice.

shown schematically in Fig. 2.3b. The transmission I_t/I_0 may be determined from the detector response with and without the sample interposed between source and detector. But α cannot be obtained from such a measurement unless reflections at the two interfaces are negligible.

2.7 REFLECTION AND TRANSMISSION AT A PLANE BOUNDARY

The considerations in the preceding section make it worthwhile to discuss reflection and transmission at plane boundaries: first, one plane boundary separating infinite media, then in the next section two successive plane boundaries forming a slab. In addition to providing useful results for bulk materials, these relatively simple boundary-value problems illustrate methods used in more complicated small-particle problems. Also, the optical properties of slabs often will be compared to those of small particles—both similarities and differences—to develop intuitive thinking about particles by way of the more familiar properties of bulk matter.

2.7.1 Normal Incidence

Consider a plane wave propagating in a nonabsorbing medium with refractive index $N_2 = n_2$, which is incident on a medium with refractive index $N_1 = n_1 + ik_1$ (Fig. 2.4). The amplitude of the incident electric field is E_i , and we assume that there are transmitted and reflected waves with amplitudes E_t and E_r , respectively. Therefore, plane-wave solutions to the Maxwell equations at